

Charlie Selmes

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Education

University of Sheffield Sheffield, UK

BSc Computer Science – Expected First

Relevant Courses: Data Structures & Algorithms, Machine Learning, Advanced Algorithms

Clubs: Ethical Hacking, Climbing, Jiu Jitsu

Sep 2023 – Jul 2026

Hills Road Sixth Form College

Cambridge, UK

A Levels: Computer Science A, Business A, Mathematics A

EPQ: A* (C# Chess game and AI)

Sep 2021 – Jun 2023

Professional Experience

BRE Group (Summer Internship – Extended)

Remote

AI Engineer

Jul 2025 – Dec 2025

- Built a **RAG-based internal knowledge assistant** using **FAISS** and an **Ollama-hosted LLM**.
- Embedded **18 company documents**, comparing retrieval approaches to improve answer relevance.
- Developed a **semantic search engine** to deliver context-aware responses from company-specific documentation.
- Presented **technical findings, limitations and recommendations** to internal stakeholders.

Self-Employed

Cambridge, UK

Software Developer

Feb 2026 – Present

- Designed an **automated video clipping platform** to convert long-form streamer content into short-form clips.
- Built **end-to-end automation scripts** to identify, process, format and prepare clips for distribution.
- Sold **50+ custom scripts and automation tools** to clients supporting creator workflow automation.
- Generated over **600,000 combined views** across clips produced using the software and automation workflows.

MyAIVA

Cambridge, UK

Software Engineer

Apr 2025 – Jul 2025

- Built an **AI-powered invoice and quote generation tool** using Travis Perkins product data.
- Integrated **WhatsApp API workflows** to support customer-facing automation and communication.
- Developed a **competitor price scraping system** to collect, compare and analyse market pricing data.
- Worked with **product data, APIs and automation pipelines** to improve quote generation efficiency.

Self-Employed

Cambridge, UK

Software Developer

Apr 2024 – Sep 2024

- Completed **23 freelance programming projects** for clients through Fiverr.
- Delivered projects across **Python, C#, C++, Java and Rust**.
- Built **automation tools, bots, web scrapers and software utilities** based on client requirements.
- Managed **client communication, requirements gathering and delivery** independently.

Technical Skills

Languages: Python, C#, C++, Java, Rust, JavaScript, SQL

Frameworks / Libraries PyTorch, TensorFlow, Selenium, FastAPI, node.js, Docker, Flask

Tools / Platforms: Git, Linux, AWS, Azure, Ollama

Other: Software engineering, client communication, problem solving, technical presentations

Selected Projects

RAG LLM on SAP Documentation – Python

Built a retrieval-augmented generation pipeline using embeddings and semantic search to answer questions from SAP documentation.

Fully Functional Chess AI – C#

Developed a playable chess engine with legal move generation, board evaluation and AI decision-making logic.

Live Video Profanity Censorship – Python

Created a real-time tool designed to detect and censor profanity in live video or audio content.

MMA TEPA Divergence Trading Bot – C++

Built a trading bot based on technical divergence signals and automated decision logic.

AI Website Penetration Testing Tool – Python (Selenium)

Created an experimental AI-assisted tool for analysing website security issues.

Social Interaction Website – Python

Created sheffmsg.fun for leaving messages around the city with over 600 users

Achievements

- LLM x Law Cambridge University Hackathon participant.
- Global Engineering Challenge Winner.
- Hack Sheffield 9 Runner Up.
- Hack Sheffield 8 Competitor.
- LeetCode Bi-Weekly Contest Competitor.

Dissertation

Disinformation Analysis Using Multi-Modal AI

Researching the use of multi-modal AI techniques to analyse and detect disinformation across different media formats.